

## **Classroom Counter Design Challenge**

Create a Classroom Counter prototype that displays the total number of people in the classroom and if the hall pass is in use.

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| <b>I Can</b>                   | <p><b>Follow the I can statement listed below to successfully complete this challenge:</b></p> <ul style="list-style-type: none"><li>• <b>I can</b> code a device to act as a counter for a DSA Classroom</li><li>• <b>I can</b> code my counter to display the total number of students in the room for that day and if the hall pass is in use</li><li>• <b>I can</b> present my prototype to my peers and DSA staff</li><li>• <b>Bonus: I can analyze and evaluate the impact of incorporating a Makey Makey.</b></li></ul>                                                                                                 |
| <b>Coding Resources</b>        | <p><b>Here are some resources you can use to help with the coding for your prototype:</b></p> <ul style="list-style-type: none"><li>• <a href="#">Microsoft Make Code</a> - Refer to the Microbit section to write any programs you need for this challenge.</li><li>• <a href="#">Scratch</a> - Here is where you can design a project and write code to be run on your Chromebook</li><li>• <b>Chromebook</b> - Your Chromebook will be a very valuable resource for writing any of your code.</li></ul>                                                                                                                     |
| <b>Materials</b>               | <p><b>Here are some materials you will have access to upon the completion of writing any code:</b></p> <ul style="list-style-type: none"><li>• Microbit</li><li>• Makey Makey</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>My Blueprint Reflection</b> | <p><b>To complete this section please log in to your My Blueprint account and select your term portfolio (the term you are currently in). Select create post, and create a journal titled "Coding Exploration". Finally copy and paste the questions below and then select post and then begin completing the questions in a medium you are comfortable with.</b></p> <ol style="list-style-type: none"><li>1. What kind of role does technology play in your life?</li><li>2. Have you ever considered a career in coding?</li><li>3. If you could code and create something what would you try to code and create?</li></ol> |

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| <b>Outcomes<br/>And<br/>Competencies</b> | <i>Learners will formulate responses to changes in Canadian society.</i><br><i>Learners will investigate technologies and careers in Science.</i><br><i>Competencies: Technological fluency, citizenship, and creativity</i> |